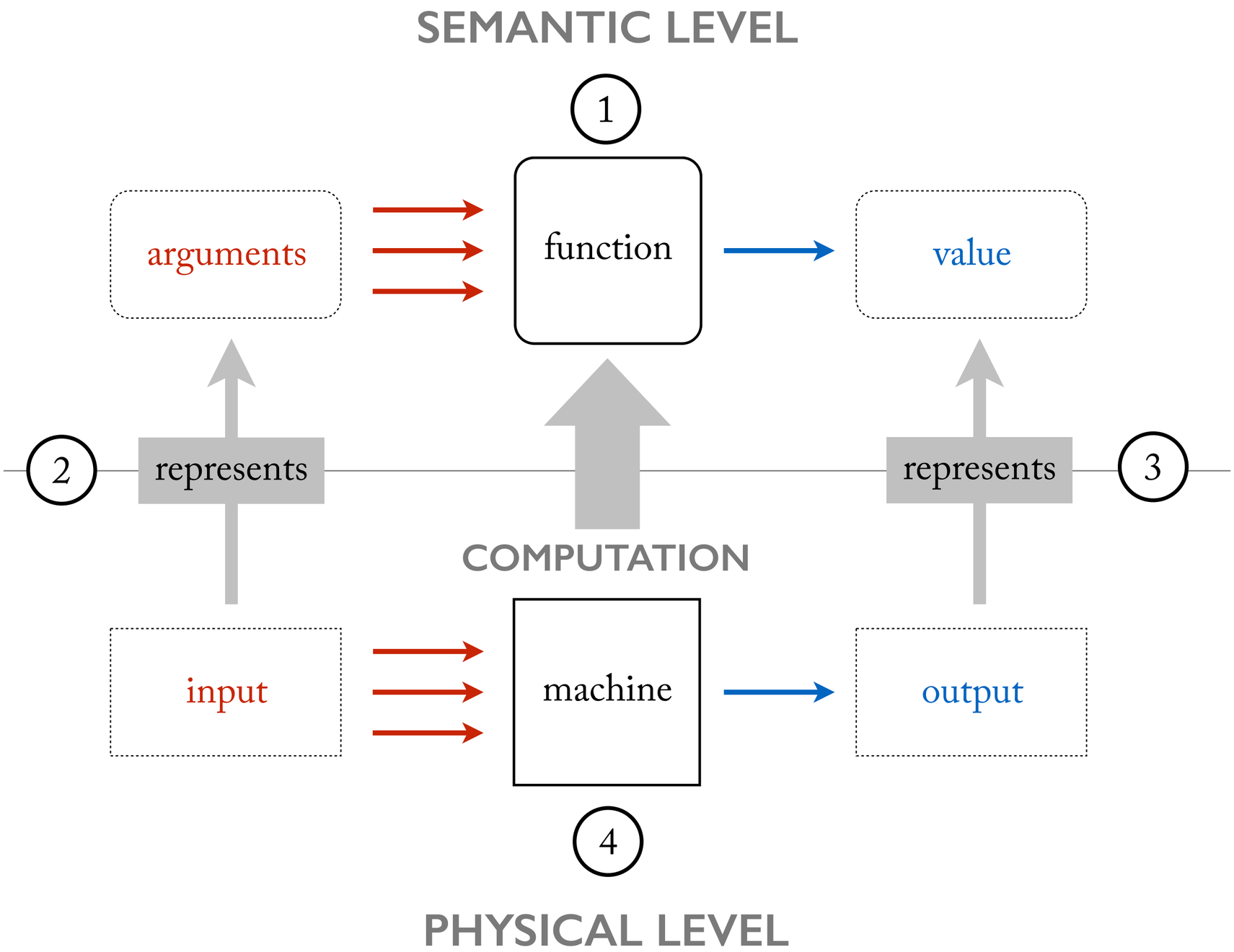
6. Defining Computation

The Structure of Computation

Computation is a relation between a **machine** and a **function** that is brought about when the machine is designed to physically **mirror** or **simulate** the abstract argue-value profile of the function. In this section, we will define this relationship formally.



Definition of Computation

Machine M **computes** function *f* under system R if and only if:

For every possible argument *x* for function *f*:

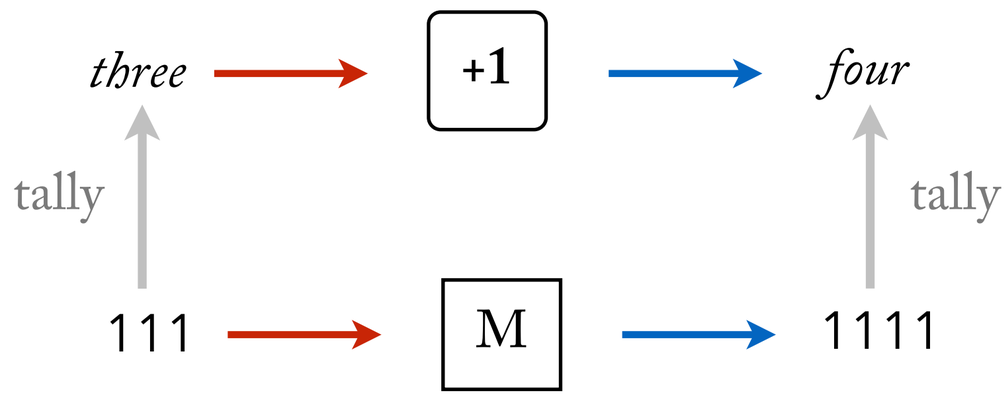
(1) there is a value *y* such that *f(x)=y* ;

(2) there is a symbol string S_1 such that S_1 represents *x* under R;

(3) there is a symbol string S_2 such that S_2 represents *y* under R;

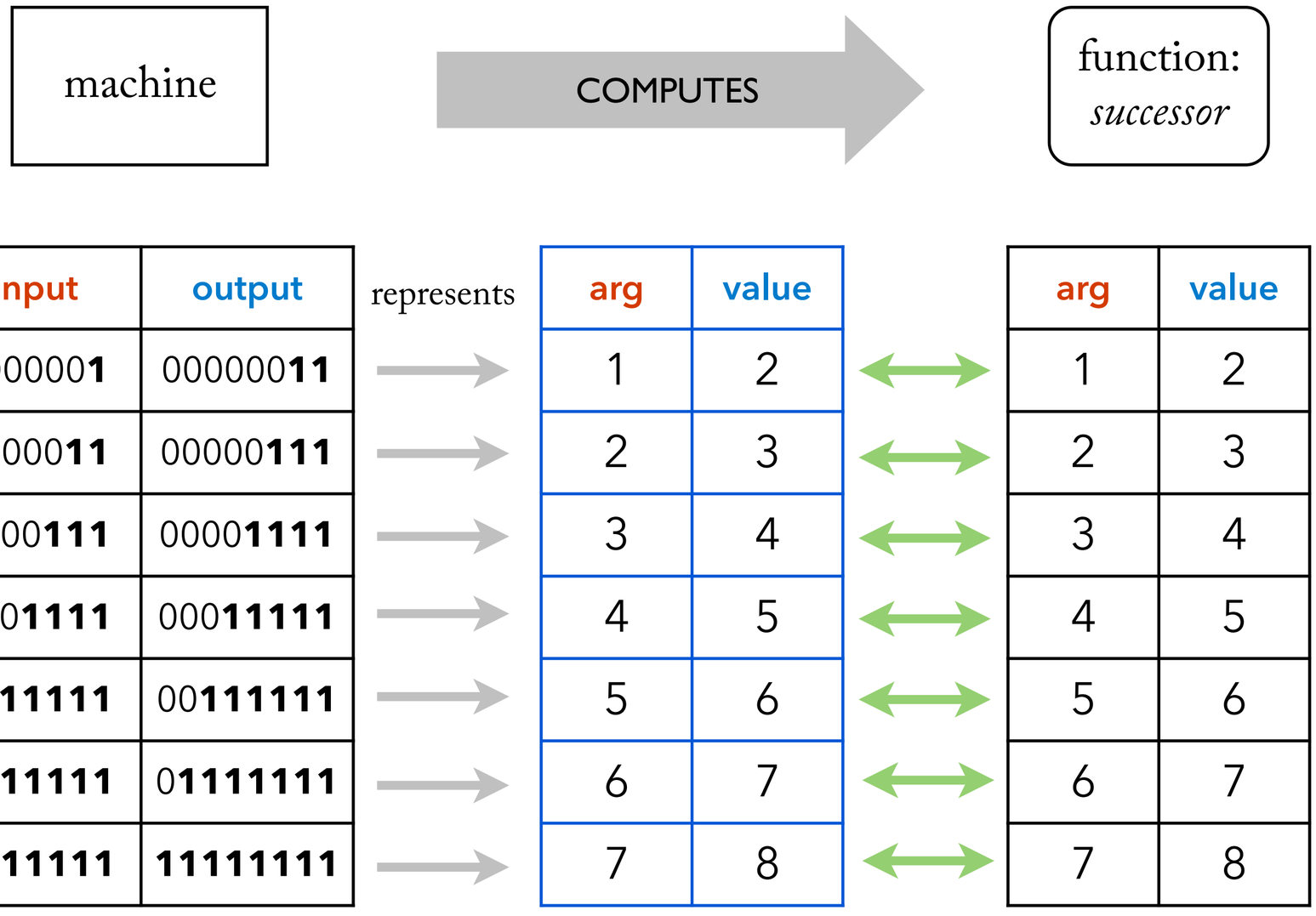
(4) if M takes S_1 as input, then M produces S_2 as output.

For at least one argument, the machine at right computes successor (+1) under the tally representation system.



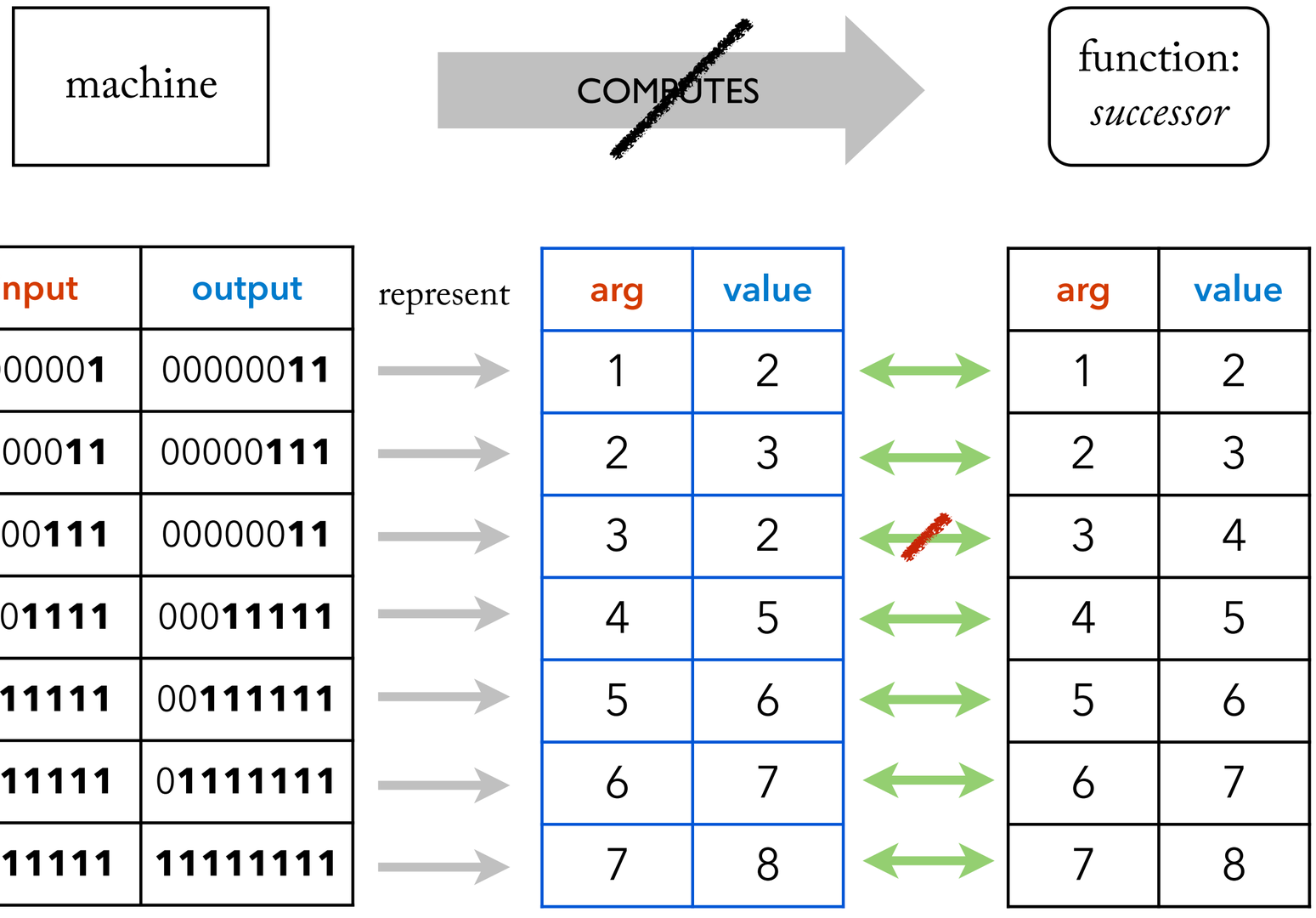
Examples of Computation

The following machine successfully computes the successor function, under the tally representation system. The machine's input-output profile, when interpreted in tally, perfectly matches the argument-value profile of the successor function (so far).



input	output	arg	value	arg	value
0000000**1**	0000000**11**	1	2	1	2
000000**11**	000000**111**	2	3	2	3
00000**111**	00000**1111**	3	4	3	4
0000**1111**	0000**11111**	4	5	4	5
000**11111**	000**111111**	5	6	5	6
00**111111**	00**1111111**	6	7	6	7
0**1111111**	0**11111111**	7	8	7	8
11111111	**111111111**				

The following machine does *not* computes the successor function under the tally representation system. The machine's input-output profile, when interpreted in tally, does *not* perfectly match the argument-value profile of the successor function.



input	output	arg	value	arg	value
0000000**1**	0000000**11**	1	2	1	2
000000**11**	000000**111**	2	3	2	3
00000**111**	00000**11**	3	2	3	4
0000**1111**	0000**11111**	4	5	4	5
000**11111**	000**111111**	5	6	5	6
00**111111**	00**1111111**	6	7	6	7
0**1111111**	0**11111111**	7	8	7	8
11111111	**111111111**				

Bounded Functions

A **bounded** function is a function which is defined for only a finite range of arguments. To name a bounded function, we write:

f[0-*n*] e.g. "*successor*[0-3]"

Note: this names a function that is only defined for the *arguments* 0 through *n* (*not* the values). The same notation applies whether the function is 1-place, 2-place, or more. For example the following two tables are equivalent to the named function in each case.

successor[0-2]

arg	val
0	1
1	2
2	3

addition[0-1]

arg1	arg2	val
0	0	0
0	1	1
1	0	1
1	1	2

A machine **computes a bounded function** when it computes that function's values for all *defined* arguments. The values it computes for undefined arguments are irrelevant for now.

Notation for Functionally Describing Computers

We will often want to refer compactly to a computer in terms of what function it computes, and what system of representation it computes it under. This is actually a deep and important form of **functional abstraction**, as we'll see in a few weeks.

I will say that a machine computes

f[0-*n*] S

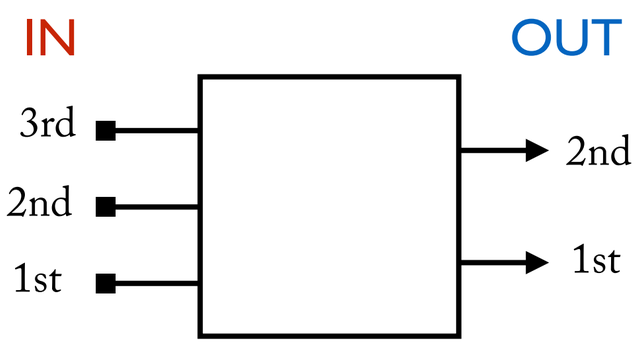
where "*f*[0-*n*]" is a (bounded) function, and S is a representational system. Here I will normally use **T** to stand for tally, and **B** to stand for binary. So I will often ask you to solve problems such as:

Design a machine which computes +1[0-9] B.
Design a machine which computes 2(*x*+3) T.

I'll sometimes omit variables in the definition of the function, if context makes their role clear, as in the first machine described above.

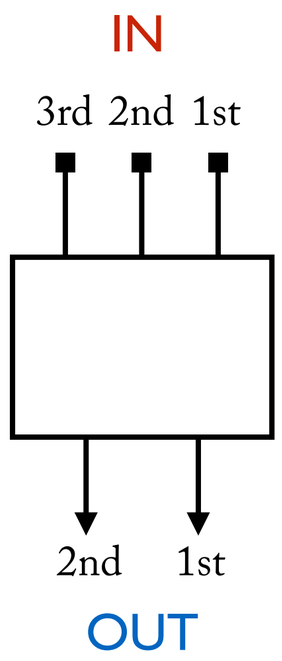
Inputs and Outputs

Input and output. A CC can have any number of inputs and any number of outputs. All the inputs are read at the same time (**synchronically**) and all the outputs are produced at the same time. By convention, the inputs and outputs are ordered from top-to-bottom, or from right-to left (following Arabic).

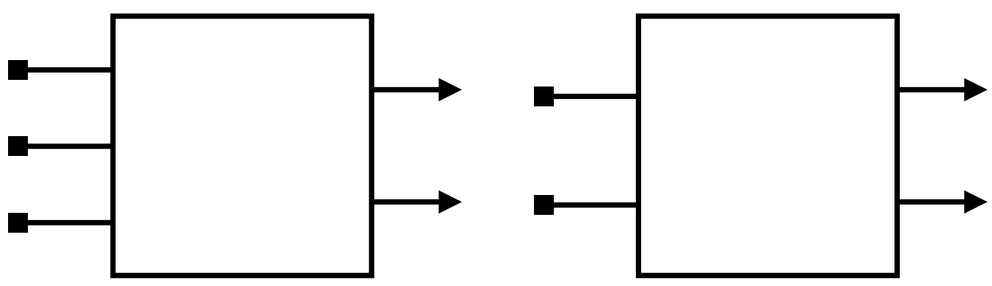


The positions here correspond to positions in a complex numeral:

1 1 0
3rd 2nd 1st



Local input. The **local** inputs to a circuit correspond to the number of input-wires the machine has. Each local input receives one symbol at a given time. A single symbol is called an **atomic** symbol.

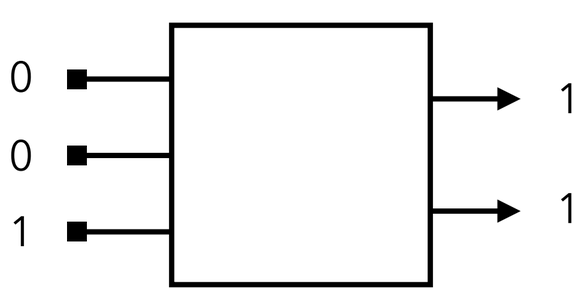


3 local inputs

2 local inputs

Global input. The **global** input to the machine is a collection of its local inputs. Where each local input carries an atomic symbol, a global input is the aggregate or **string** of atomic symbols. This string itself is called a **complex** symbol. The **order of local inputs** within a global input corresponds to the **order of symbols** in the input string.

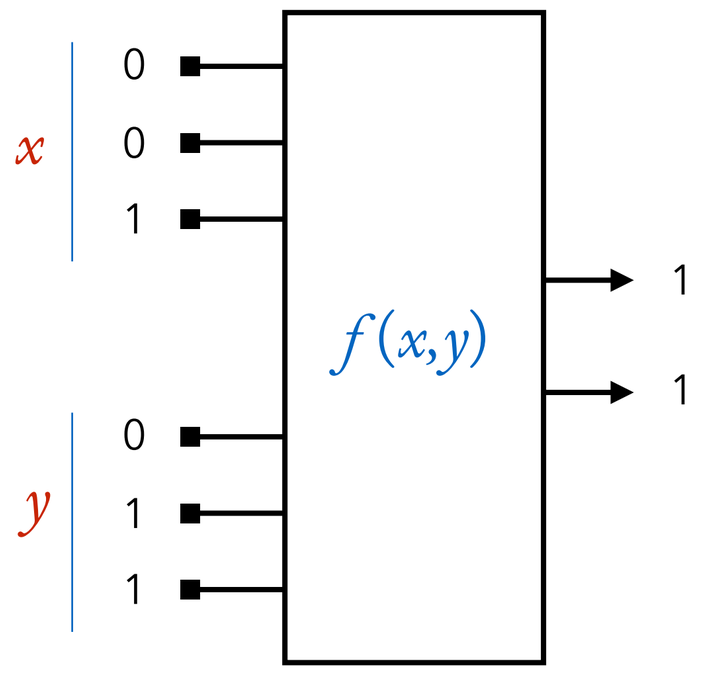
In a CC, the global input is read at one time, by reading each local input synchronically. Later we'll look at circuits where the global input is read over time, **diachronically**.



In the example on the left:
The local inputs are "1", "0", and "0".
The global input is the string "001".

Multiple global inputs. The number of global inputs corresponds to the number of arguments in the function being computed. Each global input represents one argument.

When designing a machine that computes a function with multiple argument places, indicate clearly which global input goes with which argument. The ordering convention again is top-to-bottom, or right-to-left.



Computation as a Three-Part Relation

Computation is a relation between a **machine** M, a **system of representation** R, and a **function** *f*.

Given any two, you can solve for the third.
M + R = *f*?
M + *f* = R?
f + R = M?

The machine and representational system together determined the function computed. But they are themselves logically independent from one another. This means that…

The same M can compute different functions *f* and *f** relative to different systems of representation R and R*.

The same function *f* can be computed by different machines M and M* relative to different systems of representation R and R*.

This demonstrates the importance of representational system for designing a computer. One must get the circuitry of the machine right, but it must also be tuned to the right representational system.

In-Class Problems

1. Suppose M1 computes x+1[0-4] T
    a. How many local inputs and outputs must M1 have?
    How many global inputs and outputs does M1 have?
    b. Make a global input-output table for M1. (Restrict your table to possible inputs and outputs.)
    c. Design a circuit for M1.
2. M2 computes x+1[0-5] T. Design a circuit.
3. M3 computes x+1[0-1] B. M3 has 1 local input and 2 local outputs.
    Make a global input-output table. Design a circuit.
4. M4 computes x+1[0-2] B. M3 has 2 local inputs and 2 local outputs.
    Make a global input-output table. Design a circuit.
5. M5 computes x+1[0-3] B. Make a global input-output table. Design a circuit.

Tip: for all these problems, turn your circuit 90° so the inputs and outputs are vertically aligned, the same way you would set up a long addition problem.